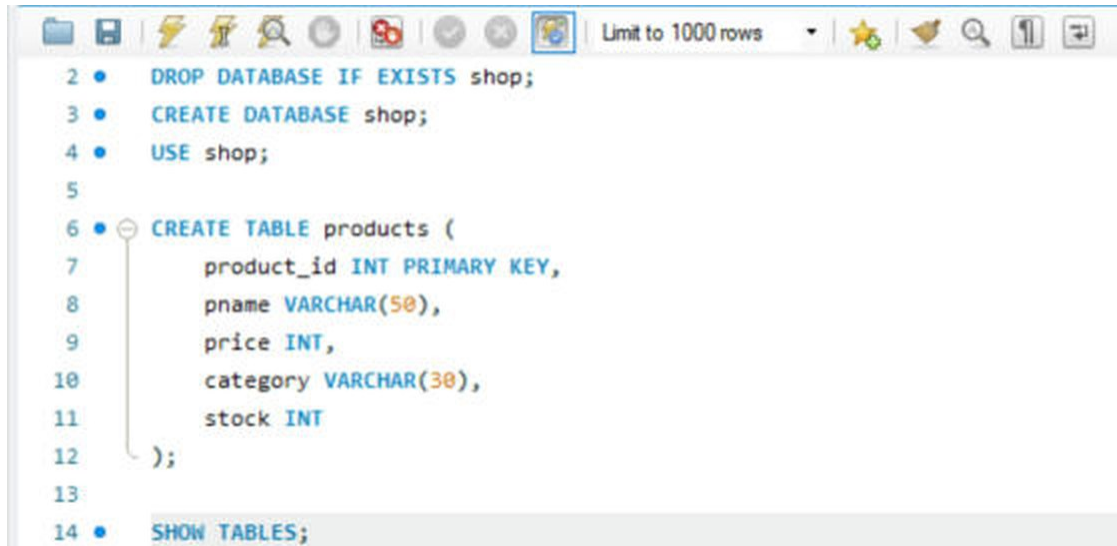


SQL Practice Set 1 – Shop Database

Practice Set 1: Create and Insert (Q1–Q5)

1. Create the shop database and the products table exactly as shown. Confirm with SHOW TABLES;.

Query (MySQL Workbench)



```
2 • DROP DATABASE IF EXISTS shop;
3 • CREATE DATABASE shop;
4 • USE shop;
5
6 • CREATE TABLE products (
7     product_id INT PRIMARY KEY,
8     pname VARCHAR(50),
9     price INT,
10    category VARCHAR(30),
11    stock INT
12 );
13
14 • SHOW TABLES;
```

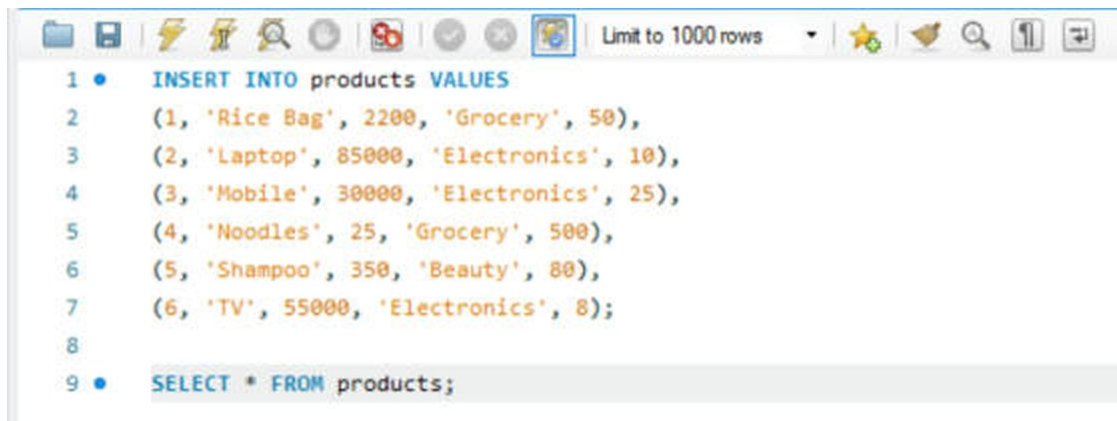
Output



Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Tables_in_shop			
products			

2. Insert all 6 products in one single INSERT statement.

Query (MySQL Workbench)



```
1 • INSERT INTO products VALUES
2     (1, 'Rice Bag', 2200, 'Grocery', 50),
3     (2, 'Laptop', 85000, 'Electronics', 10),
4     (3, 'Mobile', 30000, 'Electronics', 25),
5     (4, 'Noodles', 25, 'Grocery', 500),
6     (5, 'Shampoo', 350, 'Beauty', 80),
7     (6, 'TV', 55000, 'Electronics', 8);
8
9 • SELECT * FROM products;
```

Output

	product_id	pname	price	category	stock
▶	1	Rice Bag	2200	Grocery	50
	2	Laptop	85000	Electronics	10
	3	Mobile	30000	Electronics	25
	4	Noodles	25	Grocery	500
	5	Shampoo	350	Beauty	80
	6	TV	55000	Electronics	8

3. Insert product 7, giving only product id and pname ('Pen'). Which columns become NULL?

Query (MySQL Workbench)

```
1 • INSERT INTO products (product_id, pname) VALUES (7, 'Pen');
2
3 • SELECT * FROM products WHERE product_id = 7;
```

Output

	product_id	pname	price	category	stock
▶	7	Pen	NULL	NULL	NULL

Note: price, category and stock become NULL because no value was given for them.

4. Try to insert another product with product id = 3. Write down the exact error and explain why it happened.

Query (MySQL Workbench)

```
1 • INSERT INTO products VALUES (3, 'Headphones', 2500, 'Electronics', 15);
```

Output (error message)

#	Time	Action	Message
117	16:41:08	SELECT * FROM products WHERE product_id = 7 LIMIT 0, 1000	1 row(s) returned
118	16:41:26	INSERT INTO products VALUES (3, 'Headphones', 2500, 'Electronics', 15)	Error Code: 1062. Duplicate entry '3' for key 'products.PRIMARY'

Note: Error Code: 1062. Duplicate entry '3' for key 'products.PRIMARY'. It happened because product_id is the primary key, so it must be unique. Product id 3 (Mobile) already exists.

5. Insert two more products of your choice in one statement: one Grocery item and one Beauty item.

Query (MySQL Workbench)

```
1  INSERT INTO products VALUES
2  (8, 'Biscuits', 50, 'Grocery', 200),
3  (9, 'Face Cream', 450, 'Beauty', 60);
4
5  SELECT * FROM products;
```

Output

	product_id	pname	price	category	stock
▶	1	Rice Bag	2200	Grocery	50
	2	Laptop	85000	Electronics	10
	3	Mobile	30000	Electronics	25
	4	Noodles	25	Grocery	500
	5	Shampoo	350	Beauty	80
	6	TV	55000	Electronics	8
	7	Pen	NULL	NULL	NULL
	8	Biscuits	50	Grocery	200
	9	Face Cream	450	Beauty	60

Practice Set 1: Read (Q6–Q15)

6. Show all columns of all products.

Query (MySQL Workbench)

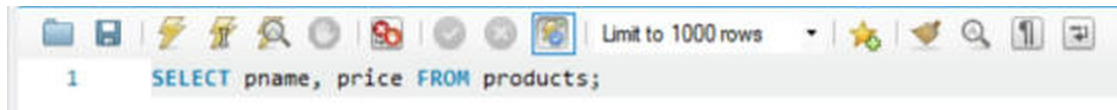
```
1  SELECT * FROM products;
```

Output

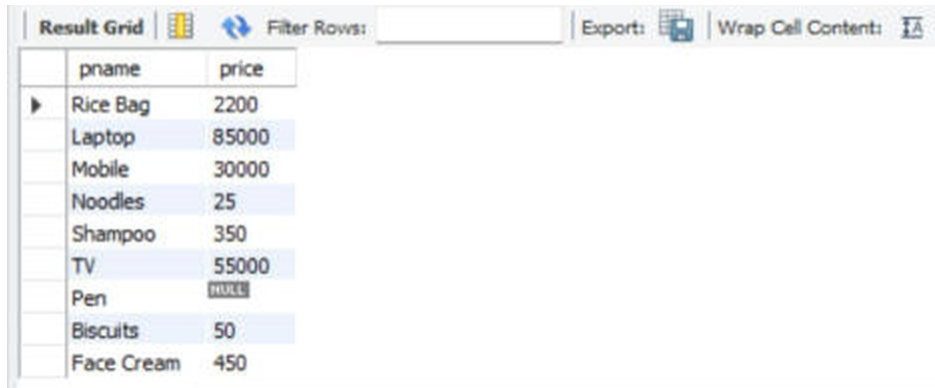
	product_id	pname	price	category	stock
▶	1	Rice Bag	2200	Grocery	50
	2	Laptop	85000	Electronics	10
	3	Mobile	30000	Electronics	25
	4	Noodles	25	Grocery	500
	5	Shampoo	350	Beauty	80
	6	TV	55000	Electronics	8
	7	Pen	NULL	NULL	NULL
	8	Biscuits	50	Grocery	200
	9	Face Cream	450	Beauty	60

7. Show only pname and price.

Query (MySQL Workbench)



Output

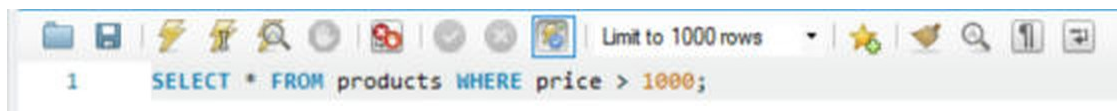


The screenshot shows the 'Result Grid' tab in MySQL Workbench. The output is a table with two columns: 'pname' and 'price'. The data rows are: Rice Bag (2200), Laptop (85000), Mobile (30000), Noodles (25), Shampoo (350), TV (55000), Pen (NULL), Biscuits (50), and Face Cream (450).

	pname	price
▶	Rice Bag	2200
	Laptop	85000
	Mobile	30000
	Noodles	25
	Shampoo	350
	TV	55000
	Pen	NULL
	Biscuits	50
	Face Cream	450

8. Show products with price above 1000.

Query (MySQL Workbench)



Output

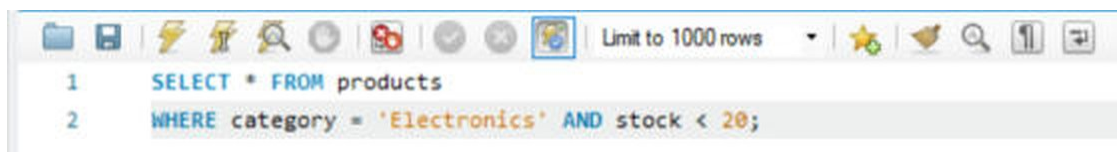


The screenshot shows the 'Result Grid' tab in MySQL Workbench. The output is a table with six columns: 'product_id', 'pname', 'price', 'category', and 'stock'. The data rows are: 1 (Rice Bag, 2200, Grocery, 50), 2 (Laptop, 85000, Electronics, 10), 3 (Mobile, 30000, Electronics, 25), and 6 (TV, 55000, Electronics, 8).

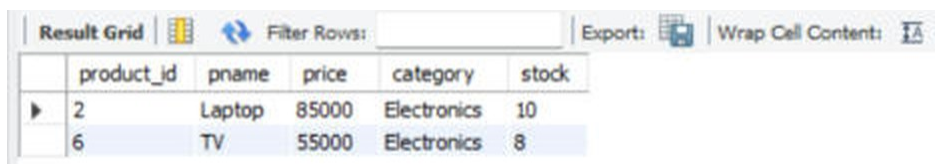
	product_id	pname	price	category	stock
▶	1	Rice Bag	2200	Grocery	50
	2	Laptop	85000	Electronics	10
	3	Mobile	30000	Electronics	25
	6	TV	55000	Electronics	8

9. Show Electronics products with stock less than 20 (AND).

Query (MySQL Workbench)



Output



The screenshot shows the 'Result Grid' tab in MySQL Workbench. The output is a table with five columns: 'product_id', 'pname', 'price', 'category', and 'stock'. The data rows are: 2 (Laptop, 85000, Electronics, 10) and 6 (TV, 55000, Electronics, 8).

	product_id	pname	price	category	stock
▶	2	Laptop	85000	Electronics	10
	6	TV	55000	Electronics	8

10. Show products that are Grocery or cost less than 500 (OR).

Query (MySQL Workbench)

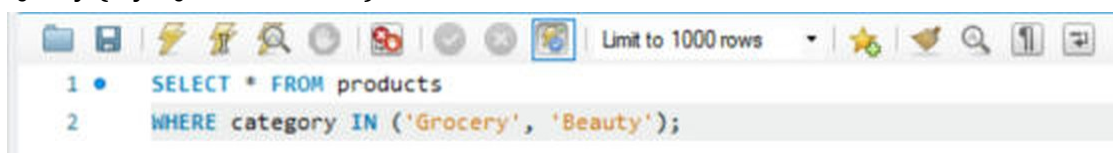


Output

	product_id	pname	price	category	stock
▶	1	Rice Bag	2200	Grocery	50
	4	Noodles	25	Grocery	500
	5	Shampoo	350	Beauty	80
	8	Biscuits	50	Grocery	200
	9	Face Cream	450	Beauty	60

11. Show products in categories Grocery and Beauty, using IN.

Query (MySQL Workbench)

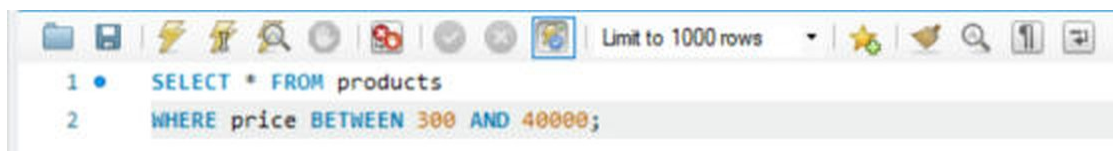


Output

	product_id	pname	price	category	stock
▶	1	Rice Bag	2200	Grocery	50
	4	Noodles	25	Grocery	500
	5	Shampoo	350	Beauty	80
	8	Biscuits	50	Grocery	200
	9	Face Cream	450	Beauty	60

12. Show products with price between 300 and 40000. Is 350 included? Why?

Query (MySQL Workbench)



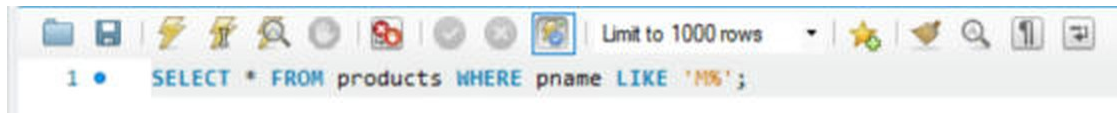
Output

	product_id	pname	price	category	stock
▶	1	Rice Bag	2200	Grocery	50
	3	Mobile	30000	Electronics	25
	5	Shampoo	350	Beauty	80
	9	Face Cream	450	Beauty	60

Note: Yes, 350 (Shampoo) is included because BETWEEN is inclusive. It counts both end values and everything in between.

13. Show products whose name starts with M.

Query (MySQL Workbench)

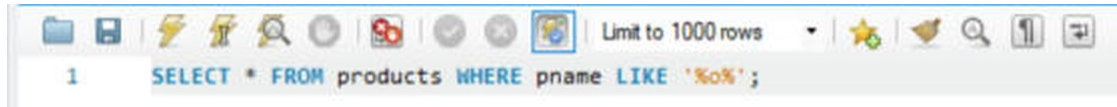


Output

product_id	pname	price	category	stock
3	Mobile	30000	Electronics	25

14. Show products whose name contains the letter o.

Query (MySQL Workbench)

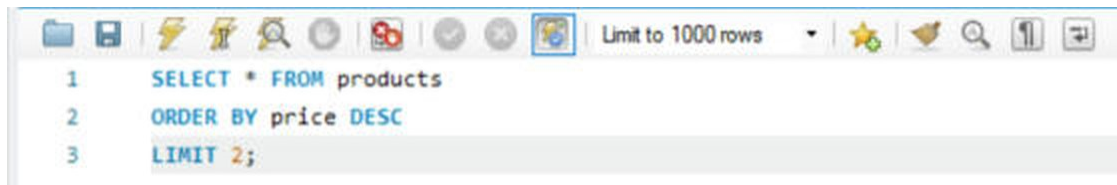


Output

product_id	pname	price	category	stock
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80

15. Show the 2 most expensive products (which two keywords together?).

Query (MySQL Workbench)



Output

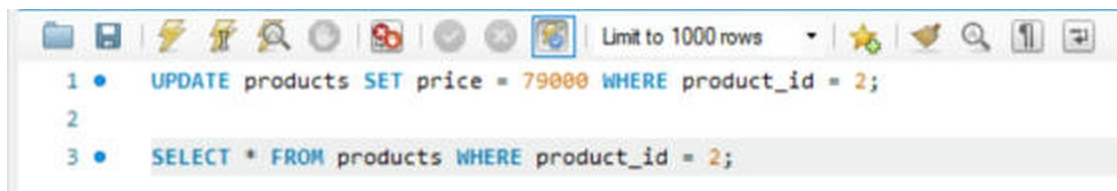
product_id	pname	price	category	stock
2	Laptop	85000	Electronics	10
6	TV	55000	Electronics	8

Note: ORDER BY price DESC and LIMIT 2 used together.

Practice Set 1: Update and Delete (Q16–Q20)

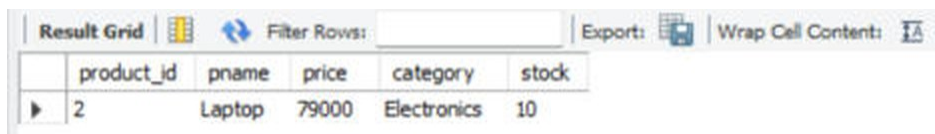
16. The Laptop price dropped to 79000. Update it using its id.

Query (MySQL Workbench)



```
1 • UPDATE products SET price = 79000 WHERE product_id = 2;
2
3 • SELECT * FROM products WHERE product_id = 2;
```

Output



product_id	pname	price	category	stock
2	Laptop	79000	Electronics	10

17. Festival offer: reduce the price of every Electronics product by 10%. Hint: SET price = price * 0.9.

Query (MySQL Workbench)



```
1 • UPDATE products
2   SET price = price * 0.9
3   WHERE category = 'Electronics';
4
5 • SELECT * FROM products WHERE category = 'Electronics';
```

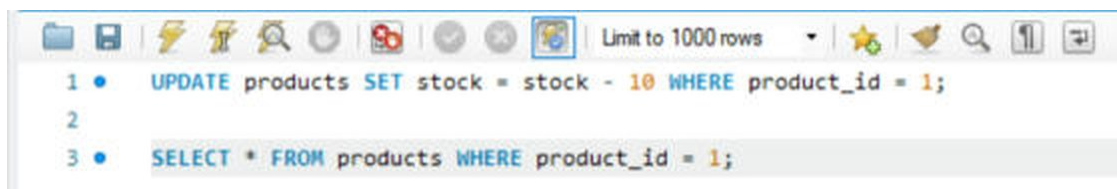
Output



product_id	pname	price	category	stock
2	Laptop	71100	Electronics	10
3	Mobile	27000	Electronics	25
6	TV	49500	Electronics	8

18. Ten Rice Bags were sold. Decrease its stock by 10 with one query.

Query (MySQL Workbench)



```
1 • UPDATE products SET stock = stock - 10 WHERE product_id = 1;
2
3 • SELECT * FROM products WHERE product_id = 1;
```

Output



product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	40

19. Delete the product named Pen. Then delete all products with stock below 10. How many rows did each query remove?

Query (MySQL Workbench)

```
1 • DELETE FROM products WHERE pname = 'Pen';
2 • DELETE FROM products WHERE stock < 10;
3
4 • SELECT * FROM products;
```

Output

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	40
2	Laptop	71100	Electronics	10
3	Mobile	27000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
8	Biscuits	50	Grocery	200
9	Face Cream	450	Beauty	60

#	Time	Action	Message
✓ 137	16:45:13	DELETE FROM products WHERE pname = 'Pen'	1 row(s) affected
✓ 138	16:45:13	DELETE FROM products WHERE stock < 10	1 row(s) affected

Note: Each query removed 1 row. The first removed Pen (id 7) and the second removed TV, the only product with stock below 10.

20. Run SELECT to check the final table. Then answer: which command would empty the table but keep it, and which would remove the table completely?

Query (MySQL Workbench)

```
1 • SELECT * FROM products;
```

Output

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	40
2	Laptop	71100	Electronics	10
3	Mobile	27000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
8	Biscuits	50	Grocery	200
9	Face Cream	450	Beauty	60

Note: TRUNCATE TABLE products; (or DELETE FROM products;) empties the table but keeps it. DROP TABLE products; removes the rows and the table completely.